

Exp no:8**Fabric Network Deployment and Management****Aim**

To deploy, manage, monitor, and maintain a Hyperledger Fabric network by creating the network, verifying network components, deploying chaincode, monitoring Docker containers, and performing network shutdown and cleanup operations.

Step 1: Navigate to the Fabric Test Network

None

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Stop any previous network

None

```
./network.sh down
```

Step 3: Start the Fabric Network

None

```
./network.sh up createChannel -ca
```

Step 4: Set the Fabric Environment

None

```
export PATH=${PWD}/../bin:$PATH
export FABRIC_CFG_PATH=$PWD/../config/

export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
```

```
export
CORE_PEER_TLS_ROOTCERT_FILE=${PWD}/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt

export
CORE_PEER_MSPCONFIGPATH=${PWD}/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp

export CORE_PEER_ADDRESS=localhost:7051

export
ORDERER_CA=${PWD}/organizations/ordererOrganizations/example.com/orderers/orderer.example.com/msp/tlscacerts/tlsca.example.com-cert.pem
```

Step 5: Verify Running Docker Containers

```
None
docker ps
```

You should see containers such as:

```
None
peer0.org1.example.com
peer0.org2.example.com
orderer.example.com
ca_org1
ca_org2
```

```

[ram@ram] ~/fabric-dev/fabric-samples/test-network
$ docker ps

```

CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
9057ff3b3af6	hyperledger/fabric-orderer:latest	orderer	"orderer"	45 seconds ago	Up 44 seconds	0.0.0.0:7050→7050/tcp, [::]:7050→7050/tcp, 0.0.0.0:7053→7053/tcp, [::]:7053→7053/tcp, 0.0.0.0:9443→9443/tcp, [::]:9443→9443/tcp
f02ea799cef8	hyperledger/fabric-peer:latest	peer0.org1.example.com	"peer node start"	45 seconds ago	Up 44 seconds	0.0.0.0:7051→7051/tcp, [::]:7051→7051/tcp, 0.0.0.0:9444→9444/tcp, [::]:9444→9444/tcp
a10d58612c52	hyperledger/fabric-peer:latest	peer0.org2.example.com	"peer node start"	45 seconds ago	Up 44 seconds	0.0.0.0:9051→9051/tcp, [::]:9051→9051/tcp, 7051/tcp, 0.0.0.0:9445→9445/tcp, [::]:9445→9445/tcp
8c4af1360d55	hyperledger/fabric-ca:latest	ca_org1	"sh -c 'fabric-ca-se...'"	50 seconds ago	Up 49 seconds	0.0.0.0:7054→7054/tcp, [::]:7054→7054/tcp, 0.0.0.0:17054→17054/tcp, [::]:17054→17054/tcp
f02f7d97b841	hyperledger/fabric-ca:latest	ca_org2	"sh -c 'fabric-ca-se...'"	50 seconds ago	Up 49 seconds	0.0.0.0:8054→8054/tcp, [::]:8054→8054/tcp, 7054/tcp, 0.0.0.0:18054→18054/tcp, [::]:18054→18054/tcp
e9c652b43459	hyperledger/fabric-ca:latest	ca_orderer	"sh -c 'fabric-ca-se...'"	50 seconds ago	Up 49 seconds	0.0.0.0:9054→9054/tcp, [::]:9054→9054/tcp, 7054/tcp, 0.0.0.0:19054→19054/tcp, [::]:19054→19054/tcp
ac23d975a119	ghcr.io/eramba/eramba:latest	cron	"/docker-cron-entryp..."	3 weeks ago	Up 4 hours	80/tcp
2e575645bc7e	ghcr.io/eramba/eramba-mcp-server:latest	mcp_server	"docker-entrypoint.s..."	3 weeks ago	Up 4 hours (healthy)	3000/tcp
f496f6dc84bd	ghcr.io/eramba/eramba:latest	eramba	"/docker-entrypoint.s..."	3 weeks ago	Up 4 hours	80/tcp
01e16bcef0b0	redis:7.4.2-alpine	redis	"docker-entrypoint.s..."	3 weeks ago	Up 4 hours	
da98fac37466	mysql:8.4.3-oracle	mysql	"docker-entrypoint.s..."	3 weeks ago	Up 4 hours	3306/tcp, 33060/tcp

Step 6: Display Docker Images

None

docker images

You should find Fabric images such as:

None

hyperledger/fabric-peer
hyperledger/fabric-orderer
hyperledger/fabric-ca

hyperledger/fabric-peer	2.5	e382f4af4538	2 months ago	155MB
hyperledger/fabric-peer	2.5.16	e382f4af4538	2 months ago	155MB
hyperledger/fabric-peer	latest	e382f4af4538	2 months ago	155MB
hyperledger/fabric-orderer	2.5	80322dee2204	2 months ago	121MB
hyperledger/fabric-orderer	2.5.16	80322dee2204	2 months ago	121MB
hyperledger/fabric-orderer	latest	80322dee2204	2 months ago	121MB
hyperledger/fabric-ccenv	2.5	85e2ca8475b0	2 months ago	675MB
hyperledger/fabric-ccenv	2.5.16	85e2ca8475b0	2 months ago	675MB
hyperledger/fabric-ccenv	latest	85e2ca8475b0	2 months ago	675MB
hyperledger/fabric-baseos	2.5	71fca91e6713	2 months ago	160MB
hyperledger/fabric-baseos	2.5.16	71fca91e6713	2 months ago	160MB
hyperledger/fabric-baseos	latest	71fca91e6713	2 months ago	160MB

Step 7: Check All Docker Containers

None

docker ps -a

This displays both running and stopped containers.

Step 8: Deploy the Asset Transfer Chaincode

```
./network.sh deployCC \
-ccl javascript \
-ccn basic \
-ccp ../asset-transfer-basic/chaincode-javascript
```

```
peer lifecycle chaincode queryinstalled
```

```
docker logs peer0.org1.example.com
```

```
docker logs --tail 10 peer0.org1.example.com
```

Step 11: Monitor Orderer Logs

None

```
docker logs orderer.example.com
```

This displays the orderer's activity.

Or, for only the latest entries:

None

```
docker logs --tail 10 orderer.example.com
```

```
(ram@ ram) - [~/fabric-dev/fabric-samples/test-network]
$ docker logs --tail 10 orderer.example.com
2026-09-08 12:56:05.919 UTC 003f WARN [orderer.common.broadcast] Handle → Error reading from 172.18.0.1:59788: rpc error: code = Canceled desc = context canceled
2026-09-08 12:56:14.879 UTC 0040 INFO [comm.grpc.server] 1 → streaming call completed grpc.service=orderer.AtomicBroadcast grpc.method=Broadcast grpc.peer_address=172.18.0.1:59788 error="rpc error: code = Canceled desc = context canceled" grpc.code=Canceled grpc.call_duration=2.92542278s
2026-09-08 12:56:14.873 UTC 0041 INFO [orderer.consensus.etcdraft] propose → Created block [4], there are 0 blocks in flight channel=mychannel node=1
2026-09-08 12:56:14.878 UTC 0042 INFO [orderer.consensus.etcdraft] writeBlock → Writing block [4] (Raft index: 6) to ledger channel=mychannel node=1
2026-09-08 12:56:14.897 UTC 0043 WARN [orderer.common.broadcast] Handle → Error reading from 172.18.0.1:40224: rpc error: code = Canceled desc = context canceled
2026-09-08 12:56:14.898 UTC 0044 INFO [comm.grpc.server] 1 → streaming call completed grpc.service=orderer.AtomicBroadcast grpc.method=Broadcast grpc.peer_address=172.18.0.1:40224 error="rpc error: code = Canceled desc = context canceled" grpc.code=Canceled grpc.call_duration=2.831526388s
2026-09-08 12:56:22.274 UTC 0045 INFO [orderer.consensus.etcdraft] propose → Created block [5], there are 0 blocks in flight channel=mychannel node=1
2026-09-08 12:56:22.278 UTC 0046 INFO [orderer.consensus.etcdraft] writeBlock → Writing block [5] (Raft index: 7) to ledger channel=mychannel node=1
2026-09-08 12:56:22.282 UTC 0047 WARN [orderer.common.broadcast] Handle → Error reading from 172.18.0.1:34408: rpc error: code = Canceled desc = context canceled
2026-09-08 12:56:22.302 UTC 0048 INFO [comm.grpc.server] 1 → streaming call completed grpc.service=orderer.AtomicBroadcast grpc.method=Broadcast grpc.peer_address=172.18.0.1:34408 error="rpc error: code = Canceled desc = context canceled" grpc.code=Canceled grpc.call_duration=2.835784398s
```

Step 12: Verify the Channel

None

```
peer channel list
```

```
(ram@ ram) - [~/fabric-dev/fabric-samples/test-network]
$ peer channel list
2026-09-08 18:28:26.984 IST 0001 INFO [channelCmd] InitCmdFactory → Endorser and orderer connections initialized
Channels peers has joined:
mychannel
```

This is optional in the manual!